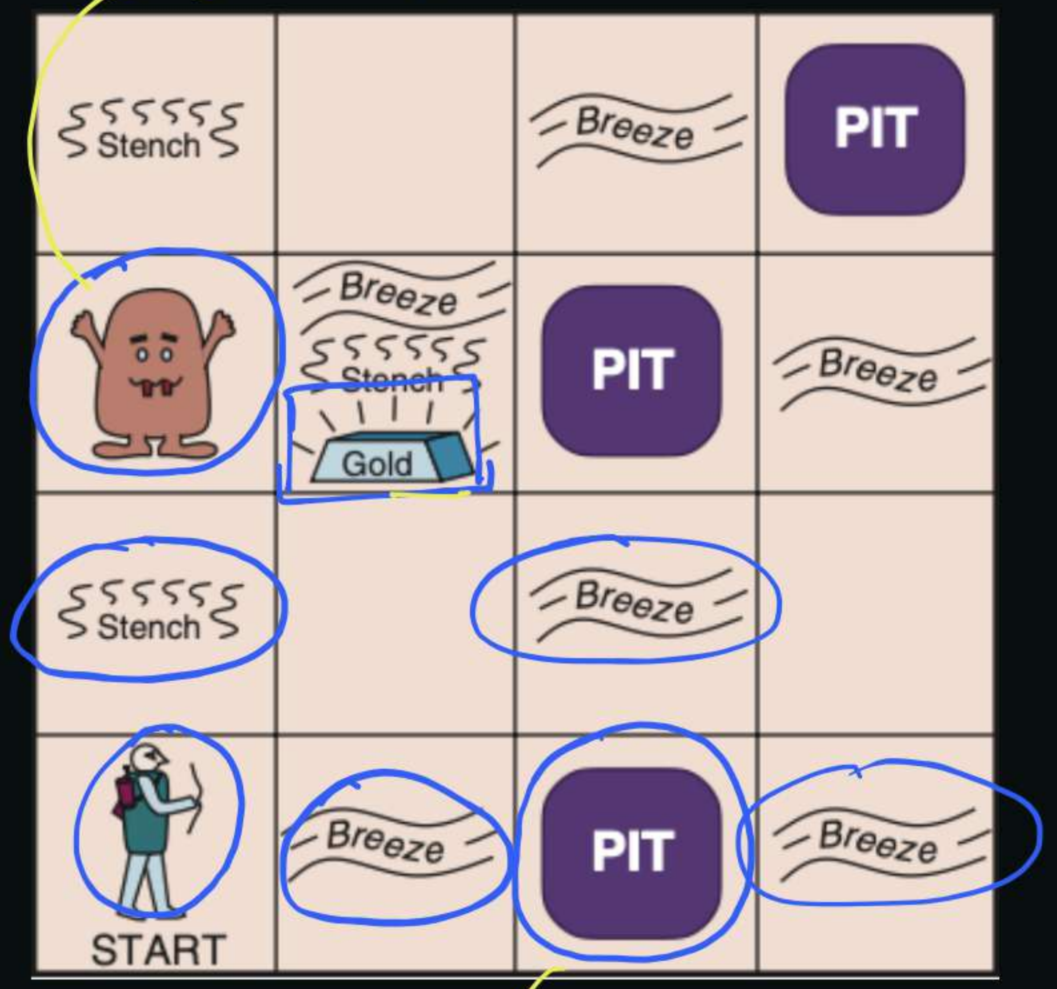
The Wumpus World :-

↳ A classic problem in AI, that demonstrates the use of knowledge representation & reasoning.

wumpus : monster

4
3
2
1



1 2 3 4

pit : hole, the agent falls & dies.

(1,1) → agent starts.

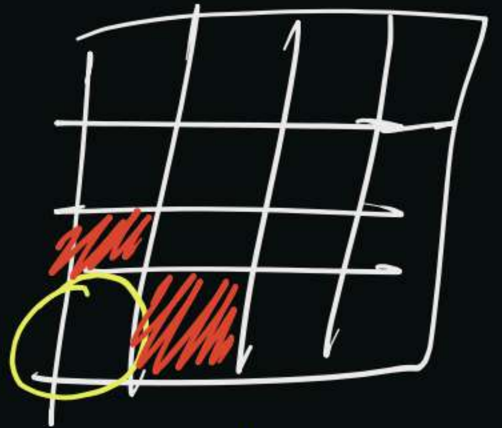
actions →
- move forward.
- turn left / right
- Shoot an arrow, to kill wumpus.
- Grab the gold.
- Climb outside the cave.

Sensory Info :
- Breeze : felt near a pit.
- Stench : near the Wumpus.
- Glitter : near the gold block.
- Scream : Heard when the Wumpus is killed.

↳ we can encode rules like, "If there is a breeze, a pit is nearby".

eg. If breeze (1,1), then pit must be in (1,2) or (2,1).

↳ (Wumpus Ahead $\wedge$ Wumpus Alive) $\Rightarrow$ Shoot.



Wumpus Ahead $\wedge$ Wumpus Alive

$\therefore$ Shoot

$$\begin{array}{l} a \rightarrow b \\ a \\ \hline \therefore\ b \end{array}$$

Horn Clauses & Definite Clauses :-

1. Horn clause :-

↳ A disjunction (OR) of literals with atmost one positive literal.

eg. $p \vee \neg q \vee \neg r$ ✓ ⟶ definite clause.

$\neg p \vee \neg q$ ✓ ⟶ negative horn clause.

$\neg p \vee q \vee r$ ✗ (not a horn clause)

horn clauses ⟶
- Definite Clause → exactly one positive literal
  eg. $p \vee \neg q$
- Negative horn clause → all negative literals
  eg. $\neg p \vee \neg q \vee \neg r$.

Forward and backward chaining :-

1. Forward Chaining :-

↳ Inference techniques, where reasoning starts from known facts and applies inference rules to derive new facts until goal is reached or no further inference is possible.

↳ Also called data driven approach.

↳ All possible conclusions are derived.

eg. Initial facts : { fever, cough, headache }

Knowledge base : (fever ^ cough) → flu

flu → take_rest

(headache ^ fever) → migraine

{fever, cough, headache}

Pass 1 :

→ (fever ∧ cough) → $\boxed{\text{flu}}$ T

→ flu? → take_rest ✗

(headache ∧ fever) → $\boxed{\text{migraine}}$ T

Inference : {flu, migraine}

Pass 2 :

{ fever, cough, headache, flu, migraine}

(fever ∧ cough) → flu

flu → $\boxed{\text{take\_rest}}$ T

(headache ∧ fever) → migraine

Inference : { take_rest}

Pass 3 : { fever, cough, headache, flu, migraine, take_rest }

Terminate → no new info can be inferred now.

2. Backward chaining :-

↳ It is a goal-driven approach, starts with a goal.

↳ Proves a specific conclusion.

eg. Knowledge base :=

(fever ∧ cough) → $\boxed{\text{flu}}$

flu → $\boxed{\text{take\_rest}}$

(headache ∧ fever) → migraine

Facts :- { fever, cough, headache }

Goal :- Prove [take_rest] → pass 1.

Prove [flu] → pass 2. bcoz. flu → take_rest.

SAT problem :-

↳ A compound proposition is satisfiable, if true for atleast 1 case.

eg. $\underbrace{(p \vee \neg q)} \wedge \underbrace{(q \vee \neg r)} \wedge \underbrace{(r \vee \neg p)}$ → product of sums.
Conjunctive Normal Form (CNF).

↳ SAT problem → check if x is satisfiable or not.

eg. $(p \vee q)$ → Is it in CNF ? Yes. $\boxed{(p \vee q) \wedge (T)}$

p → ? Yes. $(p \wedge T)$

↳ they are NP-Complete.

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Q.29 1 mark	Let x and y be two propositions. Which of the following statements **is a tautology /are tautologies**? Implies
(A)	$(\neg x \wedge y) \boxed{\Rightarrow} (y \Rightarrow x)$
(B)	$(x \wedge \neg y) \Rightarrow (\neg x \Rightarrow y)$
(C)	$(\neg x \wedge y) \Rightarrow (\neg x \Rightarrow y)$
(D)	$(x \wedge \neg y) \Rightarrow (y \Rightarrow x)$

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Q.54 2 marks

Let *game(ball, rugby)* be true if the ball is used in rugby and false otherwise.

Let *shape(ball, round)* be true if the ball is round and false otherwise.

Consider the following logical sentences:

s1: $\forall ball\ \neg\ game(ball, rugby) \Rightarrow shape(ball, round)$: $(g \vee s)$ } Same

s2: $\forall ball\ \neg\ shape(ball, round) \Rightarrow game(ball, rugby)$: $(s \vee g)$

$\boxed{s3:}$ $\forall ball\ game(ball, rugby) \Rightarrow \neg\ shape(ball, round)$: $(\neg g \vee \neg s)$ } Same.

$\boxed{s4:}$ $\forall ball\ shape(ball, round) \Rightarrow \neg\ game(ball, rugby)$: $(\neg s \vee \neg g)$

Which of the following choices is/are logical representations of the assertion,

"All balls are round except balls used in rugby"? $\forall balls\ (s \wedge \neg g)$

(A) ✓	$\boxed{s1} \wedge s3 \rightarrow (g \vee s) \wedge (\neg g \vee \neg s)$
~~(B)~~ ✗	$s1 \wedge s2 \rightarrow (g \vee s) \wedge (s \vee g) \equiv (g \vee s) \rightarrow$ All balls are round or used in rugby. ✗
(C) ✓	$\boxed{s2} \wedge s3 \rightarrow (s \vee g) \wedge (\neg g \vee \neg s)$
~~(D)~~ ✗	$s3 \wedge s4 \rightarrow (\neg g \vee \neg s) \wedge (\neg s \vee \neg g) \equiv (\neg g \vee \neg s)$

All balls are either not round or not used in rugby. ✗

$$\boxed{a \rightarrow b \equiv \neg a \vee b}$$

$$\neg s \rightarrow g \equiv \neg(\neg s) \vee g \equiv s \vee g$$

$$g \rightarrow \neg s \equiv \neg g \vee \neg s$$

$$s \rightarrow \neg g \equiv \neg s \vee \neg g$$

$$\boxed{a \vee b \equiv b \vee a}$$

(A) $\rightarrow$ $\boxed{(g \vee s) \wedge (\neg g \vee \neg s) \equiv (g \wedge \neg s) \vee (\neg g \wedge s)}$

H.W. Logic / Analytical Reasoning.

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Q.6 2 marks.	Thousands of years ago, some people began dairy farming. This coincided with a number of mutations in a particular gene that resulted in these people developing the ability to digest dairy milk. Based on the given passage, which of the following can be inferred?
(A)	All human beings can digest dairy milk.
(B)	No human being can digest dairy milk.
(C)	Digestion of dairy milk is essential for human beings.
(D)	In human beings, digestion of dairy milk resulted from a mutated gene.